

Recap.

① For set $A \subseteq \mathcal{P}(\Omega)$, $\mathcal{A}$ is generator of $\mathcal{S}(\mathcal{A}) =$ "smallest $\mathcal{S}$ -Algebra including set $\mathcal{A}$ ".

② For measurable space $(\Omega, \mathcal{A})$, $B \subseteq \Omega$ and $B \neq \emptyset$:

$\mathcal{A}|_B = \mathcal{A} \cap B = \{A \cap B \mid A \in \mathcal{A}\}$ is induced $\mathcal{S}$ -Algebra by B . (Restriction of $\mathcal{A}$ on B).

③ For $(\Omega, \mathcal{A})$, $(\Omega', \mathcal{A}')$, $X: \Omega \rightarrow \Omega'$ is measurable, if $X^{-1}(A') \in \mathcal{A}$ for $\forall A' \in \mathcal{A}'$.

④ With probability space $(\Omega, \mathcal{A}, P)$ and $\mathbb{R}$, a RV-RV is a measurable map from $\Omega \rightarrow \mathbb{R}$.

e.g. $\Omega = \{1, 2, 3\}$, RV-RV X is defined as $X(\omega) = \omega$.

⑤ Bernoulli: $f_X(x) = \begin{cases} p & x=1 \\ q=1-p & x=0 \end{cases}$ Binomial (n -times Bernoulli): $f_X(x) = \binom{n}{x} p^x q^{n-x}$

Geometric (until first success): $f_X(x) = p q^{x-1}$

Poisson: $f_X(i) = \frac{e^{-\lambda} \lambda^i}{i!}$ (λ : average speed, i : number of events)

Exercises.

Ex 1. Hint: The inverse image of a $\mathcal{S}$ -algebra is still a $\mathcal{S}$ -algebra.

(i) Left $\subseteq$ Right: $\mathcal{E}_Y \subseteq \mathcal{S}(\mathcal{E}_Y)$, so $f^{-1}(\mathcal{E}_Y) \subseteq f^{-1}(\mathcal{S}(\mathcal{E}_Y))$ which is a $\mathcal{S}$ -algebra.

$\mathcal{S}(f^{-1}(\mathcal{E}_Y))$ is the smallest $\mathcal{S}$ -algebra, must be a subset of $f^{-1}(\mathcal{S}(\mathcal{E}_Y))$.

Right $\subseteq$ Left: Define $\tilde{\mathcal{A}} = \{A' \in \mathcal{S}(\mathcal{E}_Y) \mid f^{-1}(A') \in \mathcal{S}(f^{-1}(\mathcal{E}_Y))\} \subseteq \mathcal{S}(\mathcal{E}_Y)$.

Show that $\tilde{\mathcal{A}}$ is $\mathcal{S}$ -algebra containing $\mathcal{E}_Y$: {

②③:

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Ex. 2.

② $A' \in \tilde{\mathcal{A}}$, $f^{-1}(A') \in \mathcal{S}(f^{-1}(\mathcal{E}_Y))$,

$f^{-1}(A'^c) = (f^{-1}(A'))^c \in \mathcal{S}(f^{-1}(\mathcal{E}_Y))$.

③ for sequence $\{A_i'\}$, $f^{-1}(A_i') \in \mathcal{S}(f^{-1}(\mathcal{E}_Y))$,

$f^{-1}(\bigcup_{i \in \mathbb{N}} A_i') = \bigcup_{i \in \mathbb{N}} f^{-1}(A_i') \in \mathcal{S}(f^{-1}(\mathcal{E}_Y))$.

For $\forall E \in \mathcal{E}_Y$, we have $f^{-1}(E) \in f^{-1}(\mathcal{E}_Y) \subseteq \mathcal{S}(f^{-1}(\mathcal{E}_Y)) \Rightarrow E \in \tilde{\mathcal{A}}$ and $\mathcal{S}(\mathcal{E}_Y) \subseteq \tilde{\mathcal{A}}$.

$\Rightarrow \tilde{\mathcal{A}} = \mathcal{S}(\mathcal{E}_Y)$, $f^{-1}(\mathcal{S}(\mathcal{E}_Y)) = f^{-1}(\tilde{\mathcal{A}}) \subseteq \mathcal{S}(f^{-1}(\mathcal{E}_Y))$.

(ii) " $\Rightarrow$ ": f is $\mathcal{A}_X - \mathcal{S}(\mathcal{E}_Y)$ measurable, for $\forall A' \in \mathcal{S}(\mathcal{E}_Y)$, $f^{-1}(A') \in \mathcal{A}_X$.

since $\mathcal{E}_Y \subseteq \mathcal{S}(\mathcal{E}_Y)$, then for $E' \in \mathcal{E}_Y$, $f^{-1}(E') \in \mathcal{A}_X$.

" $\Leftarrow$ ": Assume that $\forall E' \in \mathcal{E}_Y$, $f^{-1}(E') \in \mathcal{A}_X$, which means $f^{-1}(\mathcal{E}_Y) \subseteq \mathcal{A}_X$.

$\mathcal{A}_X$ is $\mathcal{S}$ -algebra, so $\mathcal{S}(f^{-1}(\mathcal{E}_Y)) \subseteq \mathcal{A}_X$, by (i) we have $f^{-1}(\mathcal{S}(\mathcal{E}_Y)) \subseteq \mathcal{A}_X$, measurable.

(iii) Not discussed, see formal solution.

Ex 2. Hint: Use the conclusion from Ex. 1.

Define function g on $A \rightarrow X$ such that $g(a) = a$ for $\forall a \in A$.

Then $g^{-1}(B) = A \cap B$ for $\forall B \subseteq X$.

$$\begin{aligned} \delta(\mathcal{E}|_A) &= \delta(\{E \cap A \mid E \in \mathcal{E}\}) = \delta(\{g^{-1}(E) \mid E \in \mathcal{E}\}) = \delta(\{g^{-1}(E)\}) \\ &\stackrel{\text{Ex. 1}}{=} g^{-1}(\delta(\mathcal{E})) = \{A \cap B \mid B \in \delta(\mathcal{E})\} = \delta(\mathcal{E})|_A. \end{aligned}$$

Ex 3. Hint: Use the approximation formula $e^{-x} = \lim_{n \rightarrow \infty} (1 - \frac{x}{n})^n$.

$$\begin{aligned} \lim_{n \rightarrow \infty} P_{\text{Bin}(n, p_n)}(k) &= \lim_{n \rightarrow \infty} \binom{n}{k} p_n^k (1-p_n)^{n-k} \\ &= \lim_{n \rightarrow \infty} \frac{n(n-1)\dots(n-k+1)}{k!} p_n^k (1-p_n)^{n-k} \\ &= \lim_{n \rightarrow \infty} \frac{n^k}{k!} p_n^k (1-p_n)^n \\ &= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} (1 - \frac{np_n}{n})^n \\ &= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} e^{-np_n} \\ &= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} e^{-\lambda}. \end{aligned}$$

Ex 4.

(a) Let event A_n be "Agathe boarded on the n -th train".

Everytime when a train leaves, # people waiting increases 1, which means when n -th train arrives, there are $n+1$ people waiting on the platform.

To "board the n -th train", Agathe has to miss first $(n-1)$ trains.

$$P(A_1) = \frac{1}{2}, \quad P(A_2) = (1 - \frac{1}{2}) \times \frac{1}{3} = \frac{1}{6}, \quad P(A_3) = (1 - \frac{1}{2}) \times (1 - \frac{1}{3}) \times \frac{1}{4} = \frac{1}{12}, \quad \dots$$

$$P(A_n) = (1 - \frac{1}{2}) \times (1 - \frac{1}{3}) \times \dots \times (1 - \frac{1}{n}) \times \frac{1}{n+1} = \frac{1}{2} \times \frac{2}{3} \times \dots \times \frac{n-1}{n} \times \frac{1}{n+1} = \frac{1}{n(n+1)}.$$

(b) Let event B be "Agathe boarded no later than the n -th train",

$$P(B) = P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i) = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{2} + \frac{1}{2} - \dots + \frac{1}{n} - \frac{1}{n+1} = \frac{n}{n+1}.$$

(Directly added because A_i 's
are independent from each other!)